Supporting Information

Accept or abstain? A pre-specified test of reliability gates calibrated on held-out chemistry for molecular property and reaction-yield models

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Contents: Section S1, results of the MLP ensemble (M2), Tables S1–S3. Section S2, operating-limits card of the random forest (M1) at $t = 0.9$ (descriptive pooled precision with correct/accepted counts), Table S4. Section S3, the analysis plan. Section S4, software environment. Section S5, per-seed results for both calibration pipelines and both targets (Tables S5–S8 for M1 and M2; Tables S9–S10 for M3). Section S6, results of the graph-network ensemble M3, added by protocol amendment 2.0 (Tables S11–S14). Table numbers in the main text refer to the main manuscript.

S1 Results of the MLP ensemble (M2)

Table S1. Ungated accuracy of M2 (mean ± s.d. over five seeds). Error rate: share of predictions with an absolute error above 0.5 log units (logP) or 10 percentage points (yield).

Test set	n	RMSE	R^2	Error rate
LogP, random R	2,449	0.62 ± 0.01	0.87 ± 0.01	0.30
LogP, held-out cluster	3,551	0.84 ± 0.02	0.74 ± 0.02	0.45
LogP, prospective E1	327	0.63 ± 0.03	0.61 ± 0.03	0.41
LogP, fluorinated E2 (all)	183	0.70 ± 0.02	-0.42 ± 0.07	0.48
sp^3 C–F compounds	140	0.70 ± 0.03	-0.70 ± 0.12	0.50
parents	43	0.70 ± 0.02	0.04 ± 0.06	0.42
BH Test1, random R	350	6.19 ± 0.45	0.95 ± 0.01	0.09
BH Test1, held-out additives	898	12.08 ± 0.95	0.80 ± 0.03	0.30
BH Test2, random R	351	6.15 ± 0.46	0.95 ± 0.00	0.10
BH Test2, held-out additives	900	12.69 ± 0.55	0.78 ± 0.02	0.34
BH Test3, random R	350	5.52 ± 0.36	0.96 ± 0.01	0.08
BH Test3, held-out additives	897	16.20 ± 0.99	0.67 ± 0.04	0.40
BH Test4, random R	350	5.47 ± 0.44	0.96 ± 0.01	0.08
BH Test4, held-out additives	900	17.87 ± 0.93	0.54 ± 0.05	0.48
SM, random R	576	10.91 ± 0.69	0.85 ± 0.02	0.25
SM, held-out ligands	960	19.48 ± 4.59	0.28 ± 0.61	0.45

Table S2. AURC of the selective error rate for M2 (lower is better; mean over seeds). "No gate" is the ungated error rate.

Held-out test	No gate	s_1	$-u$	Fused
LogP cluster	0.445	0.339	0.366	0.314
LogP prospective (E1)	0.408	0.292	0.373	0.284
LogP fluorinated (E2)	0.480	0.416	0.434	0.392
BH Test1	0.304	0.280	0.246	0.246
BH Test2	0.336	0.352	0.259	0.259
BH Test3	0.396	0.351	0.337	0.339
BH Test4	0.481	0.483	0.446	0.461
SM ligand	0.454	0.463	0.360	0.386

Table S3. Operating-limits card for Gate B with M2 at $t = 0.8$. Descriptive pooled precision with correct/accepted counts and accepting seeds for both pipelines; other definitions as in Table 4 of the main text. No independent-binomial interval is reported. The C_h-calibrated gate abstained on every query in every seed.

	C_r pipeline			C_h pipeline		
Held-out test	Coverage	Precision (correct/accepted)	Seeds	Coverage	Precision (correct/accepted)	Seeds
LogP cluster	6.87%	0.82 (1005/1222)	4/5	0.00%	abstained (0/0)	0/5
LogP prospective (E1)	8.26%	0.81 (110/135)	4/5	0.00%	abstained (0/0)	0/5
LogP fluorinated (E2)	5.68%	0.69 (36/52)	4/5	0.00%	abstained (0/0)	0/5
BH Test1	97.71%	0.70 (3084/4387)	5/5	0.00%	abstained (0/0)	0/5
BH Test2	96.58%	0.67 (2918/4346)	5/5	0.00%	abstained (0/0)	0/5
BH Test3	78.93%	0.60 (2141/3540)	4/5	0.00%	abstained (0/0)	0/5
BH Test4	99.07%	0.52 (2320/4458)	5/5	0.00%	abstained (0/0)	0/5
SM ligand	13.29%	0.60 (385/638)	1/5	0.00%	abstained (0/0)	0/5

S2 Operating-limits card of M1 at $t = 0.9$

Table S4. Operating-limits card for Gate B with M1 at $t = 0.9$. Definitions as in Table 4 of the main text.

	C_r pipeline			C_h pipeline		
Held-out test	Coverage	Precision (correct/accepted)	Seeds	Coverage	Precision (correct/accepted)	Seeds
LogP cluster	0.51%	0.91 (83/91)	2/5	0.36%	0.92 (59/64)	1/5
LogP prospective (E1)	0.80%	0.92 (12/13)	2/5	0.61%	1.00 (10/10)	1/5
LogP fluorinated (E2)	0.00%	abstained (0/0)	0/5	0.22%	1.00 (2/2)	1/5
BH Test1	45.14%	0.91 (1839/2027)	5/5	18.02%	0.94 (764/809)	3/5
BH Test2	61.04%	0.87 (2388/2747)	5/5	22.00%	0.95 (944/990)	4/5
BH Test3	34.02%	0.85 (1300/1526)	4/5	10.26%	0.95 (436/460)	3/5
BH Test4	70.49%	0.65 (2057/3172)	5/5	19.22%	0.72 (627/865)	2/5
SM ligand	26.94%	0.88 (1136/1293)	5/5	4.40%	0.90 (190/211)	1/5

S3 Analysis plan

The plan below is a historical record reproduced in its original wording, not a revised claim of statistical validity. Protocol 1.0 was committed at `f187a12`; note 1.1 at `0269f67`; pre-scoring analysis code at `feadf9a`. The timing rests on local repository history from the same working session, not an external registration. Later documentation and bootstrap corrections are recorded in note 1.2 of `PROTOCOL.md`. Notes on implementation follow the historical plan.

For the 8 November revision, statements in the original plan about a threshold guarantee are superseded by the empirical interpretation in the main manuscript. Candidates and their order use calibration data, and no finite-sample guarantee is established. The original conformal-style normalizer uses the calibration fold, its quantile convention is retained, and neither marginal coverage for this implementation nor accepted-subset coverage under shift is established. The random/held-out Gate B comparison refits the scorer and changes threshold data and sample size. The hypotheses, pass criteria and original calculations are retained.

The cards are descriptive summaries of accepted prediction–seed observations. A compound or reaction may repeat across seeds and related chemistry shares groups; the original pooled Wilson intervals were removed. Tables S5–S8 give per-seed precision, correct/accepted counts and coverage. No replacement interval or inference to unseen chemical groups/campaigns is claimed. The original H1 query-bootstrap intervals describe the retained analysis conditional on fitted pipelines, with incomplete chemical-group dependence. In logP H and SM H, partly overlapping query sets are resampled independently across seeds, so repeated-query dependence is not fully preserved. Statistical review of the target population and paired/clustered sampling design is needed before a new inferential analysis. E1 and E2 share 182 compounds; BH tests share a campaign. The eight tests are not independent replications.

S3.1 1. Question

Can an accept-or-abstain gate, calibrated on data held out *by chemistry*, keep the precision of a molecular model's accepted predictions at a stated target when the queries come from chemistry the model has not seen? And does combining an input-distance signal with a model-confidence signal rank errors better than either signal alone?

S3.2 2. Definitions fixed in advance

- **Correct prediction.** LogP: $|y - \hat{y}| \leq 0.5$ log units. Reaction yield: $|y - \hat{y}| \leq 10$ percentage points.
- **Coverage** is the share of queries the gate accepts. **Selective precision** is the share of accepted queries that are correct. **Selective error rate** = 1 − selective precision.
- **Target precisions:** $t = 0.8$ and $t = 0.9$. Risk level for the threshold guarantee: $\delta = 0.1$.

S3.3 3. Data (all public)

- **D1 LogP pool:** SangsterLogP main sheet (Zenodo 10.5281/zenodo.19387551, CC BY 4.0). Parsed with RDKit; unparsable SMILES dropped; duplicates by InChIKey merged (first record kept). Every compound whose InChIKey occurs in either external set below is removed from the pool.
- **E1, prospective external set:** the SangsterLogP "Prospective" sheet (327 compounds).
- **E2, fluorinated external set:** the 183 measured amide logP values of `blackthorn-ai/fluoricLogPpKa` (MIT), each fluorinated compound beside its non-fluorinated parent.
- **D3a Buchwald–Hartwig** (Ahneman et al. 2018; as redistributed in `SoftServeInc/yield-paper`, MIT): 3,955 reactions; the four published out-of-sample additive tests (sheets Test1–Test4, training rows 1–3057, 1–3055, 1–3058 and 1–3055 respectively) and random splits.
- **D3b Suzuki–Miyaura** (Perera et al. 2018; same redistribution): 5,760 reactions, target `Product_Yield_PCT_Area_UV`.

S3.4 4. Representations and base models

- Molecules: ECFP4 counts (radius 2, 2,048 bins) plus ten RDKit descriptors (MolWt, TPSA, NumHDonors, NumHAcceptors, NumRotatableBonds, RingCount, NumAromaticRings, FractionCSP3, HeavyAtomCount, NumHeteroatoms). Crippen MolLogP is deliberately excluded.
- Reactions: concatenated ECFP4 bit vectors (1,024 bits) of each variable component (BH: aryl halide, ligand, base, additive; SM: reactant 1, reactant 2, ligand, reagent, solvent). Component structures follow the cited redistribution.
- **M1 (primary):** random forest, 300 trees, max_features = 0.2, other settings at scikit-learn defaults. Spread u = standard deviation of the per-tree predictions.
- **M2 (secondary, robustness only):** ensemble of five multilayer perceptrons (512–128, ReLU, Adam, L2 = 1e-4, early stopping on 10% of the fit set). Spread u = standard deviation across members.
- Hyper-parameters are fixed here and are not tuned on any calibration or test fold.

S3.5 5. Splits (five seeds, 0–4)

For each dataset a pool is divided by a group label: Butina clusters of ECFP4 at Tanimoto distance 0.65 (D1), additive (D3a), ligand (D3b).

- **Held-out test H:** D1: 15% of clusters; D3a: the published TestN additives; D3b: two ligands per seed.
- **Held-out calibration C_h:** groups disjoint from H and from the fit set. D1: 15% of clusters; D3a: four additives drawn from the TestN training additives; D3b: two further ligands.
- **In-distribution part I** (all remaining groups) is split at random into the fit set F (70%), the random calibration fold C_r (15%) and the random test R (15%).
- The base model is trained on F only. Both gates (C_r-calibrated and C_h-calibrated) use the same model.
- **D1 external tests:** the D1 models are also scored on E1. A separate D1 model, trained after removing every compound with an sp^3 C–F bond (SMARTS `[CX4]F`) from the pool, is scored on E2; fluorinated compounds and parents are reported apart.

Held-out tests are therefore: D1 cluster, D1–E1, D1–E2 (D1 family); D3a Test1–Test4 and D3b ligand (D3 family).

S3.6 6. Signals

- **s1:** maximum Tanimoto similarity of the query fingerprint to F; **s5:** mean of the five largest.
- **u:** ensemble spread of the base model.
- **Conformal intervals (nominal 90%):** normalised split conformal, score $|y - \hat{y}| / (u + \varepsilon)$, $\varepsilon = 0.1 \times \text{median}(u)$ on the calibration fold.

S3.7 7. Gates

- Each calibration fold C is split at random into halves C_a and C_b.
- **Single-signal gates:** score s1, or $-u$.
- **Gate B (fused):** logistic regression of correctness on (s1, s5, log u), fitted on C_a. Its score is the predicted probability of a correct prediction.
- **Threshold by Learn-then-Test** on C_b, for every score: candidate thresholds are the score quantiles of C_b, tested in fixed sequence from the strictest that accepts at least 50 points of C_b, loosening one step at a time. A threshold passes when the binomial p-value $P(\text{Bin}(n, t) \geq k) \leq \delta$, where n points are accepted and k are correct. The loosest threshold reached before the first failure is used. If none passes, the gate abstains on every query.
- **Gate A (conjunctive, no guarantee):** accept if $s1 \geq \tau_1$ and $u \leq \tau_2$; (τ_1, τ_2) from a 20 × 20 grid of quantiles of the whole C, chosen as the largest coverage whose empirical precision on C is at least t.

S3.8 8. Metrics

Per test set, model and seed: RMSE, MAE and R^2 ungated; error rate ungated; risk–coverage curves of selective error rate and selective RMSE for each score; AURC of the selective error rate (mean over all coverage levels k/n); coverage and achieved precision of every gate at t = 0.8 and 0.9; empirical coverage of the conformal intervals on all queries and on accepted queries. Results are averaged over seeds; 95% intervals come from 1,000 bootstrap resamples of test queries, pooled over seeds.

S3.9 9. Hypotheses and pass criteria (model M1)

- **H1, the gate helps where the model is weak.** On every held-out test, the selective error rate of the fused score at 30% coverage is lower than the ungated error rate, with the 95% bootstrap interval of the difference excluding zero.
- **H2, the calibration split matters.** Over the (held-out test × target) cells, Gate B calibrated on C_r falls short of the target by more than 0.05 in at least half of the cells; calibrated on C_h, it reaches at least $t - 0.05$ in at least three quarters of the cells (seed-averaged precision).
- **H3, the combination adds something.** The AURC of Gate B's score (C_h-fitted) is at least 10% lower, relatively, than the better single signal on at least two held-out tests in each of the D1 and D3 families.
- **H4, conformal coverage under shift.** Intervals calibrated on C_r cover at least 0.88 of R; cover less than 0.85 on at least one held-out test; and, on each held-out test where they fall below 0.85, cover at least 0.88 of the queries accepted by Gate B (C_h, t = 0.8).

Every hypothesis is reported whether or not it holds.

S3.10 10. Exclusions

No client, proprietary or confidential data, code or results are used. Biological context (gene expression, perturbation or cell-line shift) and generative models are out of scope.

S3.11 Notes on implementation

- A gate that abstains on every query has no defined precision. Such seeds are left out of the seed average. A cell in which every seed abstains counts as meeting the target in H2.
- H4 cannot be checked on a test where Gate B accepts nothing; this counts as a failure of H4 for that test.
- The held-out tests for H1–H4 are the logP cluster test, E1, E2 (all 183 compounds), BH Test1–Test4 and the SM ligand test.
- Holding out four calibration additives leaves about 1,640 reactions in F for each BH test. Ungated BH accuracy is therefore not directly comparable with published results on the same tests.
- The logP held-out sets were formed by assigning whole clusters until 15% of the compounds was reached.
- The Suzuki–Miyaura component structures come from a table written for this study (`src/aoa/data.py`). The table omits spectator fragments: the internal standard, HCl and water of hydration.
- For test sets that are identical in every seed (E1, E2, BH Test1–Test4), bootstrap intervals use the same resample for all seeds.

S3.12 Amendment 2.0, 12 November 2026: a graph network as a third learner (M3)

Fixed before any M3 prediction was computed. Sections 1–10 and notes 1.1–1.4 are not changed, and the verdicts of version 1, which M1 decides, stand.

Why. Under version 1 the outcome of H3 differed by learner. With M1, the fused score ranked errors no better than ensemble spread on any held-out test (relative AURC reduction between −0.9% and +1.7%). With M2, similarity s1 ranked logP errors better than spread on all three logP tests, and the fused score lowered the AURC there by 2.8–7.4%, short of the 10% criterion; on the reaction tests spread was the better signal and fusion added nothing. Two fingerprint learners cannot show whether H3 depends on the model family. A directed message-passing neural network (D-MPNN) learns its own molecular representation and is the family most uncertainty benchmarks for molecular property prediction use. This difference was seen after version 1 was scored, so the amendment is motivated post hoc; every M3 setting and criterion below is fixed before M3 exists.

M3, a graph network.

- *Software.* Chemprop 2.3.1 (PyTorch 2.15.0, Lightning 2.6.6, RDKit 2026.03.6), in a separate environment because it requires an older RDKit (versions in `env/graph-requirements.txt`). Data, splits, s1 and s5, gates, intervals and metrics are computed in the main environment exactly as for M1 and M2 (`scripts/graph_jobs.py`; the M3 fit is `scripts/graph_train.py`).
- *Input.* D1: the SMILES of the compound. D3a and D3b: the SMILES of the variable components of section 4, joined by "." into one multi-fragment graph; an empty component (no ligand or no reagent in D3b) is left out.
- *Architecture and training.* Chemprop's defaults, not tuned on any fold: bond-based message passing (hidden size 300, depth 3, no dropout), norm aggregation, a feed-forward head with one hidden layer of 300, standardised targets, mean-squared-error loss, batch size 64 and 50 epochs, with the learning rate rising from 1e-4 to 1e-3 over two warm-up epochs and decaying to 1e-4. A random 10% of F selects the checkpoint with the lowest validation loss; the other 90% trains. Training uses the CPU only, three threads per job.
- *Ensemble.* Five members. Member m of seed s uses the random seed $100 \cdot s + m$ for its initialisation, data order and validation subset. $\hat{y}$ is the member mean; u is the standard deviation across members (population, as for M2).
- *Everything else* is as in sections 3–8: M3 is trained on F only, with the same five seeds and splits, and its predictions pass through the same gates and metrics.

Hypotheses for M3. H1–H4 of section 9 are applied to M3 with unchanged criteria and read with notes 1.1, 1.3 and 1.4. The question that motivated this amendment is H3; no direction is predicted. M3's verdicts are reported beside M1's and M2's.

Descriptive, not tested. For each held-out test: which single signal, s1 or −u, has the lower AURC under M3, and the fused score's relative AURC change, beside M1 and M2; and the ungated accuracy of M3 beside M1 and M2. Reporting follows notes 1.3 and 1.4 (per-seed counts; pooled precision is descriptive, with no interval).

Failures. A member whose predictions are not finite is retrained once with the seed 100·s + m + 50. If that also fails, the ensemble keeps its other members and the tables say so. Every task and seed is reported.

Order. This amendment and the code that produces M3 predictions are committed before any M3 prediction is computed. Before this amendment was written, a timing check trained D-MPNN members for two epochs on 1,000 fit-set rows from three job files (D1, D3a Test1 and D3b, seed 0), and a smoke test trained a five-member ensemble on 300 fit-set rows of the D3a Test1 job (seed 0) and predicted 50 further fit-set rows, twice, with identical results. Neither computed a prediction on any calibration, test or external set.

2.1, 12 November 2026, after M3 was scored: documentation only. Nothing in amendment 2.0 changes.

- *Training.* All 35 task-seed fits completed; every one of the 175 members gave finite predictions, so the retry rule was not used. 103 members reached their lowest validation loss in the last three of the 50 epochs (median epoch 48).
- *Where the results are.* M3 predictions are in `results/predictions_m3/` and its analysis in `results/amendment2/`, apart from version 1's files, which stay byte-identical (`results/revision/baseline_manifest.json`). This separation was made before the M3 analysis was run.
- *Exploratory.* Written after M3 was scored and not part of the amendment: the ratio of mean spread on each held-out test to that on R, and the mean width of the C_r-calibrated intervals, for each learner (`scripts/exploratory_m3.py`).

S4 Software environment

Python 3.12.14; RDKit 2026.09.1; scikit-learn 1.9.1; NumPy 2.5.4; SciPy 1.18.1; pandas 3.1.0; matplotlib 3.11.2. The locked dependencies are installed with `uv sync --frozen --python 3.12`. The full pipeline (`fetch_data.py`, `run_predictions.py`, `analyse.py`, `make_figures.py`, `make_tables.py`) runs in about 40 minutes on an 18-core laptop CPU. Input digests (SHA-256) are listed in `PROVENANCE.md` and checked by `fetch_data.py`. M3 (amendment 2.0) runs in a separate environment, because Chemprop requires an older RDKit: Chemprop 2.3.1, PyTorch 2.15.0, Lightning 2.6.6, RDKit 2026.03.6 (`env/graph-requirements.txt`). Its 35 fits (seven tasks, five seeds, five members each) ran on the CPU, six jobs at a time with three threads each (`scripts/graph_jobs.py`); the main environment computes the splits, the similarity signals, the gates and the metrics as for M1 and M2.

S5 Per-seed descriptive operating results

These tables extend the descriptive reporting in this revision; they do not change the pre-specified hypothesis criteria. Coverage includes zero-acceptance seeds. Undefined precision is not replaced by zero or one. Counts are correct/accepted predictions within each seed; repeated queries across seeds remain paired observations, not independent trials. The machine-readable files `results/revision/gate_per_seed.csv` and `gate_pooled_descriptive.csv` also give test counts and family labels. H2 retains its original convention that an all-abstaining cell meets the target vacuously.

Table S5. Per-seed Gate B results, RF, $t = 0.8$. Coverage is the percentage of test queries accepted. A dash denotes undefined precision when no query is accepted; counts are retained.

Test / seed	C_r pipeline			C_h pipeline		
	Coverage	Correct/accepted	Precision	Coverage	Correct/accepted	Precision
D1: LogP cluster / 0	14.03%	398/500	0.796	0.00%	0/0	—
D1: LogP cluster / 1	0.48%	13/17	0.765	3.63%	110/129	0.853
D1: LogP cluster / 2	12.10%	359/425	0.845	6.95%	209/244	0.857
D1: LogP cluster / 3	22.09%	580/777	0.746	12.08%	332/425	0.781
D1: LogP cluster / 4	14.43%	426/521	0.818	0.00%	0/0	—
D1: LogP prospective (E1) / 0	21.10%	54/69	0.783	0.00%	0/0	—
D1: LogP prospective (E1) / 1	0.31%	1/1	1.000	5.50%	18/18	1.000
D1: LogP prospective (E1) / 2	16.51%	39/54	0.722	11.93%	29/39	0.744
D1: LogP prospective (E1) / 3	38.23%	95/125	0.760	22.94%	62/75	0.827
D1: LogP prospective (E1) / 4	17.74%	49/58	0.845	0.00%	0/0	—
D1: LogP fluorinated (E2) / 0	12.57%	13/23	0.565	0.00%	0/0	—
D1: LogP fluorinated (E2) / 1	0.00%	0/0	—	2.73%	3/5	0.600
D1: LogP fluorinated (E2) / 2	12.02%	17/22	0.773	4.92%	6/9	0.667
D1: LogP fluorinated (E2) / 3	25.14%	36/46	0.783	10.93%	15/20	0.750
D1: LogP fluorinated (E2) / 4	8.20%	10/15	0.667	0.00%	0/0	—
D3: BH Test1 / 0	95.21%	660/855	0.772	0.00%	0/0	—
D3: BH Test1 / 1	92.43%	616/830	0.742	53.01%	419/476	0.880
D3: BH Test1 / 2	92.87%	630/834	0.755	42.09%	347/378	0.918
D3: BH Test1 / 3	100.00%	702/898	0.782	70.60%	550/634	0.868
D3: BH Test1 / 4	89.20%	573/801	0.715	52.67%	395/473	0.835
D3: BH Test2 / 0	99.33%	651/894	0.728	57.56%	470/518	0.907
D3: BH Test2 / 1	96.89%	660/872	0.757	97.11%	660/874	0.755
D3: BH Test2 / 2	95.89%	657/863	0.761	53.00%	428/477	0.897
D3: BH Test2 / 3	99.67%	678/897	0.756	36.22%	306/326	0.939
D3: BH Test2 / 4	99.33%	630/894	0.705	66.11%	511/595	0.859
D3: BH Test3 / 0	99.89%	539/896	0.602	0.00%	0/0	—
D3: BH Test3 / 1	98.44%	551/883	0.624	27.87%	227/250	0.908
D3: BH Test3 / 2	100.00%	605/897	0.674	14.05%	124/126	0.984
D3: BH Test3 / 3	100.00%	586/897	0.653	75.36%	519/676	0.768
D3: BH Test3 / 4	89.41%	548/802	0.683	26.53%	221/238	0.929
D3: BH Test4 / 0	97.11%	560/874	0.641	33.22%	249/299	0.833
D3: BH Test4 / 1	100.00%	519/900	0.577	72.56%	413/653	0.632
D3: BH Test4 / 2	97.78%	517/880	0.588	99.89%	520/899	0.578
D3: BH Test4 / 3	100.00%	514/900	0.571	66.67%	391/600	0.652
D3: BH Test4 / 4	100.00%	527/900	0.586	77.78%	437/700	0.624
D3: SM ligand / 0	43.75%	267/420	0.636	44.06%	265/423	0.626
D3: SM ligand / 1	36.77%	283/353	0.802	0.00%	0/0	—
D3: SM ligand / 2	44.38%	325/426	0.763	27.19%	239/261	0.916
D3: SM ligand / 3	50.73%	346/487	0.710	0.00%	0/0	—
D3: SM ligand / 4	48.23%	352/463	0.760	0.00%	0/0	—

Table S6. Per-seed Gate B results, RF, $t = 0.9$. Coverage is the percentage of test queries accepted. A dash denotes undefined precision when no query is accepted; counts are retained.

Test / seed	C_r pipeline			C_h pipeline		
	Coverage	Correct/accepted	Precision	Coverage	Correct/accepted	Precision
D1: LogP cluster / 0	1.26%	43/45	0.956	0.00%	0/0	—
D1: LogP cluster / 1	0.00%	0/0	—	1.80%	59/64	0.922
D1: LogP cluster / 2	1.31%	40/46	0.870	0.00%	0/0	—
D1: LogP cluster / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 4	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 0	1.53%	5/5	1.000	0.00%	0/0	—
D1: LogP prospective (E1) / 1	0.00%	0/0	—	3.06%	10/10	1.000
D1: LogP prospective (E1) / 2	2.45%	7/8	0.875	0.00%	0/0	—
D1: LogP prospective (E1) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 4	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 0	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 3	0.00%	0/0	—	1.09%	2/2	1.000
D1: LogP fluorinated (E2) / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 0	64.92%	513/583	0.880	0.00%	0/0	—
D3: BH Test1 / 1	30.07%	255/270	0.944	0.00%	0/0	—
D3: BH Test1 / 2	37.86%	315/340	0.926	18.49%	164/166	0.988
D3: BH Test1 / 3	55.57%	457/499	0.916	42.43%	359/381	0.942
D3: BH Test1 / 4	37.31%	299/335	0.893	29.18%	241/262	0.920
D3: BH Test2 / 0	47.00%	394/423	0.931	23.78%	209/214	0.977
D3: BH Test2 / 1	68.33%	541/615	0.880	0.00%	0/0	—
D3: BH Test2 / 2	54.56%	438/491	0.892	24.22%	213/218	0.977
D3: BH Test2 / 3	89.67%	637/807	0.789	24.67%	209/222	0.941
D3: BH Test2 / 4	45.67%	378/411	0.920	37.33%	313/336	0.932
D3: BH Test3 / 0	43.26%	330/388	0.851	0.00%	0/0	—
D3: BH Test3 / 1	52.17%	375/468	0.801	9.92%	87/89	0.978
D3: BH Test3 / 2	46.49%	362/417	0.868	0.00%	0/0	—
D3: BH Test3 / 3	0.00%	0/0	—	32.11%	267/288	0.927
D3: BH Test3 / 4	28.21%	233/253	0.921	9.25%	82/83	0.988
D3: BH Test4 / 0	52.67%	350/474	0.738	0.00%	0/0	—
D3: BH Test4 / 1	84.11%	460/757	0.608	0.00%	0/0	—
D3: BH Test4 / 2	52.67%	345/474	0.728	48.89%	324/440	0.736
D3: BH Test4 / 3	71.89%	413/647	0.638	0.00%	0/0	—
D3: BH Test4 / 4	91.11%	489/820	0.596	47.22%	303/425	0.713
D3: SM ligand / 0	28.44%	227/273	0.832	21.98%	190/211	0.900
D3: SM ligand / 1	21.77%	174/209	0.833	0.00%	0/0	—
D3: SM ligand / 2	31.35%	267/301	0.887	0.00%	0/0	—
D3: SM ligand / 3	22.19%	199/213	0.934	0.00%	0/0	—
D3: SM ligand / 4	30.94%	269/297	0.906	0.00%	0/0	—

Table S7. Per-seed Gate B results, MLP, $t = 0.8$. Coverage is the percentage of test queries accepted. A dash denotes undefined precision when no query is accepted; counts are retained.

Test / seed	C_r pipeline			C_h pipeline		
	Coverage	Correct/accepted	Precision	Coverage	Correct/accepted	Precision
D1: LogP cluster / 0	9.82%	287/350	0.820	0.00%	0/0	—
D1: LogP cluster / 1	3.94%	119/140	0.850	0.00%	0/0	—
D1: LogP cluster / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 3	11.94%	349/420	0.831	0.00%	0/0	—
D1: LogP cluster / 4	8.64%	250/312	0.801	0.00%	0/0	—
D1: LogP prospective (E1) / 0	12.23%	33/40	0.825	0.00%	0/0	—
D1: LogP prospective (E1) / 1	4.28%	11/14	0.786	0.00%	0/0	—
D1: LogP prospective (E1) / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 3	13.76%	39/45	0.867	0.00%	0/0	—
D1: LogP prospective (E1) / 4	11.01%	27/36	0.750	0.00%	0/0	—
D1: LogP fluorinated (E2) / 0	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 1	10.93%	13/20	0.650	0.00%	0/0	—
D1: LogP fluorinated (E2) / 2	8.74%	10/16	0.625	0.00%	0/0	—
D1: LogP fluorinated (E2) / 3	5.46%	7/10	0.700	0.00%	0/0	—
D1: LogP fluorinated (E2) / 4	3.28%	6/6	1.000	0.00%	0/0	—
D3: BH Test1 / 0	94.65%	621/850	0.731	0.00%	0/0	—
D3: BH Test1 / 1	99.78%	659/896	0.735	0.00%	0/0	—
D3: BH Test1 / 2	99.78%	613/896	0.684	0.00%	0/0	—
D3: BH Test1 / 3	99.78%	638/896	0.712	0.00%	0/0	—
D3: BH Test1 / 4	94.54%	553/849	0.651	0.00%	0/0	—
D3: BH Test2 / 0	90.89%	568/818	0.694	0.00%	0/0	—
D3: BH Test2 / 1	97.78%	605/880	0.688	0.00%	0/0	—
D3: BH Test2 / 2	96.89%	591/872	0.678	0.00%	0/0	—
D3: BH Test2 / 3	98.89%	568/890	0.638	0.00%	0/0	—
D3: BH Test2 / 4	98.44%	586/886	0.661	0.00%	0/0	—
D3: BH Test3 / 0	96.88%	524/869	0.603	0.00%	0/0	—
D3: BH Test3 / 1	99.22%	553/890	0.621	0.00%	0/0	—
D3: BH Test3 / 2	99.55%	528/893	0.591	0.00%	0/0	—
D3: BH Test3 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 4	99.00%	536/888	0.604	0.00%	0/0	—
D3: BH Test4 / 0	97.89%	508/881	0.577	0.00%	0/0	—
D3: BH Test4 / 1	98.11%	453/883	0.513	0.00%	0/0	—
D3: BH Test4 / 2	99.56%	438/896	0.489	0.00%	0/0	—
D3: BH Test4 / 3	100.00%	474/900	0.527	0.00%	0/0	—
D3: BH Test4 / 4	99.78%	447/898	0.498	0.00%	0/0	—
D3: SM ligand / 0	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 1	66.46%	385/638	0.603	0.00%	0/0	—
D3: SM ligand / 2	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 3	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 4	0.00%	0/0	—	0.00%	0/0	—

Table S8. Per-seed Gate B results, MLP, $t = 0.9$. Coverage is the percentage of test queries accepted. A dash denotes undefined precision when no query is accepted; counts are retained.

Test / seed	C_r pipeline			C_h pipeline		
	Coverage	Correct/accepted	Precision	Coverage	Correct/accepted	Precision
D1: LogP cluster / 0	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 4	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 0	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 4	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 0	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 0	94.65%	621/850	0.731	0.00%	0/0	—
D3: BH Test1 / 1	99.78%	659/896	0.735	0.00%	0/0	—
D3: BH Test1 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 3	82.52%	537/741	0.725	0.00%	0/0	—
D3: BH Test1 / 4	54.68%	343/491	0.699	0.00%	0/0	—
D3: BH Test2 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 4	99.00%	536/888	0.604	0.00%	0/0	—
D3: BH Test4 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 1	98.11%	453/883	0.513	0.00%	0/0	—
D3: BH Test4 / 2	48.22%	243/434	0.560	0.00%	0/0	—
D3: BH Test4 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 4	99.78%	447/898	0.498	0.00%	0/0	—
D3: SM ligand / 0	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 1	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 2	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 3	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 4	0.00%	0/0	—	0.00%	0/0	—

Table S9. Per-seed Gate B results, DMPNN, $t = 0.8$. Coverage is the percentage of test queries accepted. A dash denotes undefined precision when no query is accepted; counts are retained.

Test / seed	C_r pipeline			C_h pipeline		
	Coverage	Correct/accepted	Precision	Coverage	Correct/accepted	Precision
D1: LogP cluster / 0	69.69%	1770/2483	0.713	0.00%	0/0	—
D1: LogP cluster / 1	36.01%	933/1279	0.729	0.00%	0/0	—
D1: LogP cluster / 2	25.88%	736/909	0.810	3.36%	99/118	0.839
D1: LogP cluster / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 4	0.00%	0/0	—	44.89%	1225/1621	0.756
D1: LogP prospective (E1) / 0	70.95%	189/232	0.815	0.00%	0/0	—
D1: LogP prospective (E1) / 1	43.73%	130/143	0.909	0.00%	0/0	—
D1: LogP prospective (E1) / 2	33.64%	94/110	0.855	7.65%	21/25	0.840
D1: LogP prospective (E1) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 4	0.00%	0/0	—	48.62%	140/159	0.881
D1: LogP fluorinated (E2) / 0	34.97%	56/64	0.875	7.10%	12/13	0.923
D1: LogP fluorinated (E2) / 1	3.83%	5/7	0.714	0.00%	0/0	—
D1: LogP fluorinated (E2) / 2	1.09%	2/2	1.000	3.28%	5/6	0.833
D1: LogP fluorinated (E2) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 4	11.48%	19/21	0.905	0.00%	0/0	—
D3: BH Test1 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 4	0.00%	0/0	—	27.67%	200/249	0.803
D3: BH Test3 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 3	1.90%	15/17	0.882	0.00%	0/0	—
D3: BH Test3 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 0	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 1	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 2	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 3	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 4	0.00%	0/0	—	0.00%	0/0	—

Table S10. Per-seed Gate B results, DMPNN, $t = 0.9$. Coverage is the percentage of test queries accepted. A dash denotes undefined precision when no query is accepted; counts are retained.

Test / seed	C_r pipeline			C_h pipeline		
	Coverage	Correct/accepted	Precision	Coverage	Correct/accepted	Precision
D1: LogP cluster / 0	1.26%	41/45	0.911	0.00%	0/0	—
D1: LogP cluster / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP cluster / 4	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 0	2.45%	8/8	1.000	0.00%	0/0	—
D1: LogP prospective (E1) / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP prospective (E1) / 4	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 0	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 1	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 2	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 3	0.00%	0/0	—	0.00%	0/0	—
D1: LogP fluorinated (E2) / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test1 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test2 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test3 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 0	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 1	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 2	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 3	0.00%	0/0	—	0.00%	0/0	—
D3: BH Test4 / 4	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 0	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 1	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 2	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 3	0.00%	0/0	—	0.00%	0/0	—
D3: SM ligand / 4	0.00%	0/0	—	0.00%	0/0	—

S6 Results of the graph-network ensemble (M3)

M3, an ensemble of five Chemprop D-MPNNs, was added by protocol amendment 2.0, which was committed before any M3 prediction was computed (Section S3). Its splits, similarity signals, gates and metrics are those of M1 and M2.

Table S11. Ungated accuracy of M3 (mean ± s.d. over five seeds). Definitions as in Table S1.

Test set	n	RMSE	R^2	Error rate
LogP, random R	2,449	0.57 ± 0.02	0.89 ± 0.00	0.26
LogP, held-out cluster	3,551	0.72 ± 0.04	0.81 ± 0.02	0.35
LogP, prospective E1	327	0.45 ± 0.05	0.80 ± 0.05	0.23
LogP, fluorinated E2 (all)	183	0.44 ± 0.04	0.43 ± 0.10	0.26
sp^3 C–F compounds	140	0.44 ± 0.05	0.31 ± 0.15	0.26
parents	43	0.44 ± 0.04	0.63 ± 0.07	0.25
BH Test1, random R	350	12.00 ± 1.02	0.81 ± 0.03	0.35
BH Test1, held-out additives	898	16.56 ± 0.97	0.63 ± 0.04	0.45
BH Test2, random R	351	13.06 ± 1.05	0.78 ± 0.03	0.39
BH Test2, held-out additives	900	14.05 ± 0.19	0.73 ± 0.01	0.42
BH Test3, random R	350	10.90 ± 0.57	0.83 ± 0.02	0.32
BH Test3, held-out additives	897	18.33 ± 0.91	0.57 ± 0.04	0.43
BH Test4, random R	350	12.19 ± 0.73	0.80 ± 0.02	0.35
BH Test4, held-out additives	900	21.17 ± 1.84	0.36 ± 0.11	0.58
SM, random R	576	16.96 ± 1.03	0.64 ± 0.04	0.52
SM, held-out ligands	960	22.34 ± 2.86	0.14 ± 0.42	0.62

Table S12. AURC of the selective error rate for M3 (lower is better; mean over seeds). "No gate" is the ungated error rate.

Held-out test	No gate	s_1	$-u$	Fused
LogP cluster	0.354	0.298	0.260	0.252
LogP prospective (E1)	0.234	0.178	0.148	0.142
LogP fluorinated (E2)	0.261	0.214	0.229	0.213
BH Test1	0.448	0.425	0.329	0.327
BH Test2	0.418	0.429	0.273	0.328
BH Test3	0.433	0.426	0.321	0.321
BH Test4	0.582	0.562	0.436	0.435
SM ligand	0.617	0.612	0.600	0.612

Table S13. Operating-limits card for Gate B with M3 at $t = 0.8$. Descriptive pooled precision with correct/accepted counts and accepting seeds for both pipelines; other definitions as in Table 4 of the main text. No independent-binomial interval is reported.

	C_r pipeline			C_h pipeline		
Held-out test	Coverage	Precision (correct/accepted)	Seeds	Coverage	Precision (correct/accepted)	Seeds
LogP cluster	26.31%	0.74 (3439/4671)	3/5	9.65%	0.76 (1324/1739)	2/5
LogP prospective (E1)	29.66%	0.85 (413/485)	3/5	11.25%	0.88 (161/184)	2/5
LogP fluorinated (E2)	10.27%	0.87 (82/94)	4/5	2.08%	0.89 (17/19)	2/5
BH Test1	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
BH Test2	0.00%	abstained (0/0)	0/5	5.53%	0.80 (200/249)	1/5
BH Test3	0.38%	0.88 (15/17)	1/5	0.00%	abstained (0/0)	0/5
BH Test4	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
SM ligand	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5

Table S14. Operating-limits card for Gate B with M3 at $t = 0.9$. Definitions as in Table S13.

	C_r pipeline			C_h pipeline		
Held-out test	Coverage	Precision (correct/accepted)	Seeds	Coverage	Precision (correct/accepted)	Seeds
LogP cluster	0.25%	0.91 (41/45)	1/5	0.00%	abstained (0/0)	0/5
LogP prospective (E1)	0.49%	1.00 (8/8)	1/5	0.00%	abstained (0/0)	0/5
LogP fluorinated (E2)	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
BH Test1	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
BH Test2	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
BH Test3	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
BH Test4	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5
SM ligand	0.00%	abstained (0/0)	0/5	0.00%	abstained (0/0)	0/5